

# PROS6-A1640-OSB

Serie PROS · R513A · senza serbatoio / without tank



PROS6-A1640-OSB

RESA FRIGORIFERA · COOLING CAPACITY

**1.077** kW

EER · 12/7 °C · AIR 35 °C

**2,96**

RESA NOMINALE · 15 °C · AIR 25 °C

**1.637** kW

PORTATA ACQUA · WATER FLOW

**185,3** m<sup>3</sup>/h

|                            |                 |                                        |                        |
|----------------------------|-----------------|----------------------------------------|------------------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | 13.600 × 2.050 × 2.400 |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —                      |
| Ventilatori / Fans         | 24              | Taglia / Size                          | 1640                   |
| Refrigerante / Refrigerant | R513A · GWP 631 | Campo aria esterna / Ambient air range | -5 / +50 °C            |

## Resa in sintesi / Capacity at a glance <sup>kW</sup>

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C | 50 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 1.201 | 1.140 | 1.077 | 1.013 | 947   | 881   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 1.637 | 1.558 | 1.476 | 1.393 | 1.307 | 1.220 |

Refrigeratori centralizzati condensati ad aria con compressori a vite, per le grandi potenze industriali e commerciali, con parzializzazione della potenza 25-50-75-100 %.

Centralised air-cooled chillers with screw compressors for large industrial and commercial loads, with 25-50-75-100 % capacity steps.

# Dati tecnici

## Technical data

PROS6 - A1640 - OSB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.637,26 kW              |
| Resa frigorifera / Cooling capacity | 1.408.044 kcal/h         |
| Potenza assorbita / Power input     | 309,47 kW                |
| EER / EER                           | 5,29                     |
| Portata acqua / Water flow          | 281,61 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.077,07 kW              |
| Potenza assorbita / Power input     | 363,59 kW                |
| EER / EER                           | 2,96                     |
| ESEER / ESEER                       | 3,94                     |
| Portata acqua / Water flow          | 185,26 m <sup>3</sup> /h |

### Componenti / Components

|                                             |                      |
|---------------------------------------------|----------------------|
| Compressori / Compressors                   | 2 × Vite             |
| Circuiti frigoriferi / Refrigerant circuits | 2                    |
| Ventilatori / Fans                          | 24                   |
| Condensatore / Condenser                    | a microcanali        |
| Evaporatore / Evaporator                    | a fascio tubiero     |
| Refrigerante / Refrigerant                  | R513A • GWP 631 • A1 |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 13.600 × 2.050 × 2.400 mm |
| Campo aria esterna / Ambient air range                                                    | -5 / +50 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 • 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 1.055,5 kW   |
| MEG 30 % | × 0,950 | 1.023,2 kW   |
| MEG 40 % | × 0,920 | 990,9 kW     |
| MPG 20 % | × 0,975 | 1.050,1 kW   |
| MPG 30 % | × 0,960 | 1.034,0 kW   |
| MPG 40 % | × 0,950 | 1.023,2 kW   |

# Prestazioni Performance

PROS6-A1640-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30    | 35    | 40    | 45    | 50    |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1             | 1.031 | 984   | 937   | 887   | 836   | 785   | 732   | 678   |
| 2             | 1.075 | 1.027 | 978   | 927   | 874   | 820   | 765   | 709   |
| 3             | 1.120 | 1.070 | 1.019 | 966   | 912   | 856   | 799   | 741   |
| 4             | 1.168 | 1.116 | 1.063 | 1.008 | 952   | 894   | 835   | 775   |
| 5             | 1.215 | 1.162 | 1.107 | 1.050 | 992   | 932   | 871   | 808   |
| 6             | 1.266 | 1.211 | 1.154 | 1.095 | 1.034 | 972   | 909   | 844   |
| 7             | 1.316 | 1.260 | 1.201 | 1.140 | 1.077 | 1.013 | 947   | 881   |
| 8             | 1.370 | 1.311 | 1.250 | 1.187 | 1.122 | 1.056 | 988   | 919   |
| 9             | 1.424 | 1.363 | 1.300 | 1.235 | 1.168 | 1.099 | 1.029 | 957   |
| 10            | 1.481 | 1.418 | 1.353 | 1.286 | 1.216 | 1.145 | 1.072 | 998   |
| 11            | 1.539 | 1.473 | 1.406 | 1.336 | 1.265 | 1.191 | 1.116 | 1.040 |
| 12            | 1.599 | 1.532 | 1.462 | 1.390 | 1.316 | 1.240 | 1.162 | 1.083 |
| 13            | 1.660 | 1.590 | 1.518 | 1.444 | 1.367 | 1.289 | 1.209 | 1.127 |
| 14            | 1.724 | 1.652 | 1.578 | 1.501 | 1.422 | 1.341 | 1.258 | 1.173 |
| 15            | 1.789 | 1.714 | 1.637 | 1.558 | 1.476 | 1.393 | 1.307 | 1.220 |
| 16            | 1.857 | 1.780 | 1.700 | 1.619 | 1.534 | 1.448 | 1.359 | 1.269 |
| 17            | 1.925 | 1.845 | 1.763 | 1.679 | 1.592 | 1.503 | 1.412 | 1.319 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| 1             | 247 | 271 | 297 | 326 | 357 | 392 | 431 | 473 |
| 2             | 248 | 272 | 298 | 327 | 358 | 394 | 432 | 475 |
| 3             | 249 | 273 | 299 | 328 | 360 | 395 | 434 | 477 |
| 4             | 250 | 274 | 300 | 328 | 361 | 396 | 435 | 478 |
| 5             | 251 | 274 | 301 | 329 | 361 | 397 | 436 | 479 |
| 6             | 252 | 275 | 301 | 330 | 363 | 398 | 437 | 481 |
| 7             | 253 | 276 | 302 | 331 | 364 | 399 | 439 | 482 |
| 8             | 254 | 277 | 303 | 332 | 365 | 400 | 440 | 484 |
| 9             | 254 | 278 | 304 | 333 | 366 | 401 | 441 | 485 |
| 10            | 255 | 279 | 305 | 334 | 366 | 402 | 442 | 486 |
| 11            | 256 | 280 | 306 | 335 | 367 | 403 | 443 | 487 |
| 12            | 257 | 281 | 307 | 336 | 368 | 404 | 444 | 488 |
| 13            | 258 | 281 | 308 | 337 | 369 | 405 | 445 | 489 |
| 14            | 259 | 283 | 309 | 337 | 370 | 406 | 446 | 491 |
| 15            | 261 | 284 | 309 | 338 | 371 | 407 | 447 | 492 |
| 16            | 262 | 285 | 310 | 339 | 372 | 408 | 448 | 493 |
| 17            | 263 | 286 | 311 | 340 | 373 | 409 | 449 | 494 |

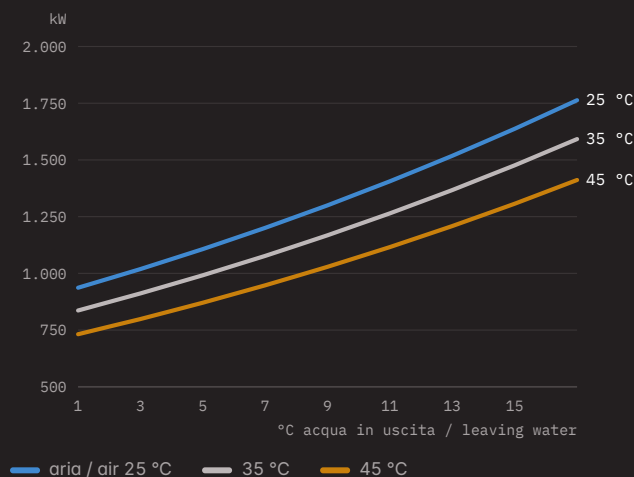
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   | 50   |
|---------------|------|------|------|------|------|------|------|------|
| 1             | 4,17 | 3,63 | 3,15 | 2,72 | 2,34 | 2,00 | 1,70 | 1,43 |
| 2             | 4,33 | 3,78 | 3,28 | 2,84 | 2,44 | 2,08 | 1,77 | 1,49 |
| 3             | 4,49 | 3,92 | 3,41 | 2,95 | 2,54 | 2,17 | 1,84 | 1,56 |
| 4             | 4,67 | 4,08 | 3,55 | 3,07 | 2,64 | 2,26 | 1,92 | 1,62 |
| 5             | 4,84 | 4,23 | 3,68 | 3,19 | 2,74 | 2,35 | 2,00 | 1,69 |
| 6             | 5,03 | 4,40 | 3,83 | 3,31 | 2,85 | 2,44 | 2,08 | 1,76 |
| 7             | 5,21 | 4,56 | 3,97 | 3,44 | 2,96 | 2,54 | 2,16 | 1,83 |
| 8             | 5,40 | 4,73 | 4,13 | 3,57 | 3,08 | 2,64 | 2,25 | 1,90 |
| 9             | 5,60 | 4,91 | 4,28 | 3,71 | 3,20 | 2,74 | 2,33 | 1,97 |
| 10            | 5,80 | 5,09 | 4,44 | 3,85 | 3,32 | 2,85 | 2,43 | 2,05 |
| 11            | 6,00 | 5,27 | 4,60 | 3,99 | 3,44 | 2,95 | 2,52 | 2,13 |
| 12            | 6,21 | 5,46 | 4,77 | 4,14 | 3,57 | 3,07 | 2,62 | 2,22 |
| 13            | 6,43 | 5,65 | 4,94 | 4,29 | 3,70 | 3,18 | 2,71 | 2,30 |
| 14            | 6,65 | 5,85 | 5,11 | 4,45 | 3,84 | 3,30 | 2,82 | 2,39 |
| 15            | 6,86 | 6,05 | 5,29 | 4,61 | 3,98 | 3,42 | 2,92 | 2,48 |
| 16            | 7,09 | 6,25 | 5,48 | 4,77 | 4,13 | 3,55 | 3,03 | 2,58 |
| 17            | 7,32 | 6,46 | 5,66 | 4,93 | 4,27 | 3,68 | 3,14 | 2,67 |

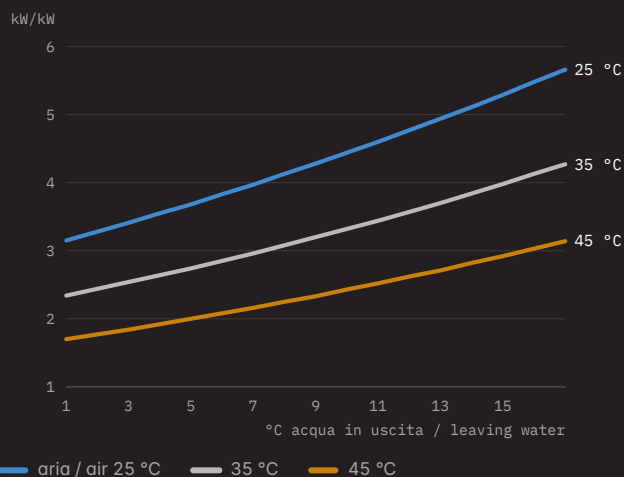
# Curve e dimensioni

## Curves and dimensions

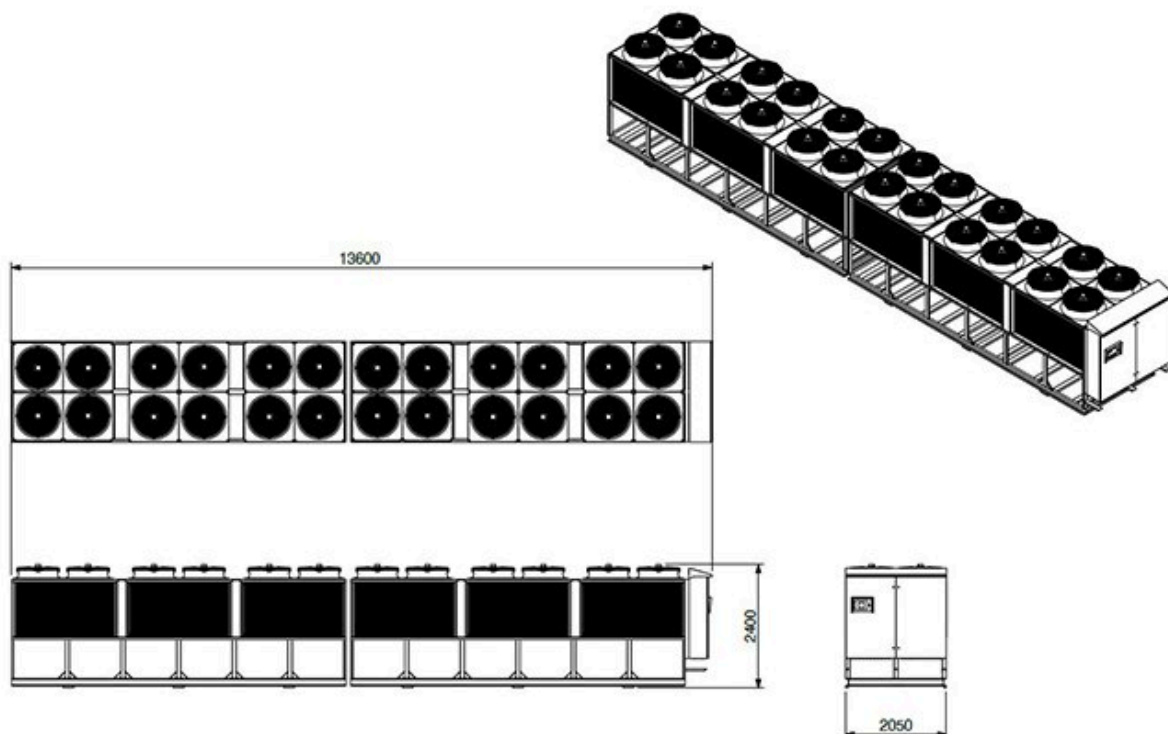
### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 13.600 × 2.050 × 2.400 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

Optional — 1/2

Options

PROS6 - A1640 - OSB · 56 optional disponibili per questa configurazione · 56 options available for this configuration

| Pompe / Pumps                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pompa 195 m³/h – prevalenza 25 m<br>Pump 195 m³/h – head 25 m                                                                                                                               |
| Pompa 195 m³/h – prevalenza 40 m<br>Pump 195 m³/h – head 40 m                                                                                                                               |
| Pompa doppia montata 195 m³/h – prevalenza 25 m<br>Twin pump, factory -fitted, 195 m³/h – head 25 m                                                                                         |
| Pompa doppia montata 195 m³/h – prevalenza 40 m<br>Twin pump, factory -fitted, 195 m³/h – head 40 m                                                                                         |
| Serbatoi e vasi di espansione / Tanks and expansion vessels                                                                                                                                 |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory -fitted twin AISI 304 pump                         |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory -fitted twin AISI 304 pump                         |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata<br>External water tank 3000 l, AISI 304 with factory -fitted twin pump                                              |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata<br>External water tank 5000 l, AISI 304 with factory -fitted twin pump                                              |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory -fitted single AISI 304 pump                      |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory -fitted single AISI 304 pump                      |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata<br>External water tank 3000 l, AISI 304 with factory -fitted single pump                                           |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata<br>External water tank 5000 l, AISI 304 with factory -fitted single pump                                           |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 3000 l, pressurised, galvanised steel with factory -fitted single pump |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 5000 l, pressurised, galvanised steel with factory -fitted single pump |
| Vaso di espansione 100 l<br>Expansion vessel 100 l                                                                                                                                          |
| Vaso di espansione 200 l<br>Expansion vessel 200 l                                                                                                                                          |
| Vaso di espansione 300 l<br>Expansion vessel 300 l                                                                                                                                          |
| Vaso di espansione 500 l<br>Expansion vessel 500 l                                                                                                                                          |
| Vaso di espansione 750 l<br>Expansion vessel 750 l                                                                                                                                          |

| Valvole e filtri / Valves and filters                    |
|----------------------------------------------------------|
| Elettrovalvola 2"<br>Solenoid valve 2"                   |
| Elettrovalvola 3"<br>Solenoid valve 3"                   |
| Elettrovalvola 4"<br>Solenoid valve 4"                   |
| Elettrovalvola 5"<br>Solenoid valve 5"                   |
| Elettrovalvola 6"<br>Solenoid valve 6"                   |
| Filtro a dischi in acciaio 2"<br>Steel disc filter 2"    |
| Filtro a dischi in acciaio 3"<br>Steel disc filter 3"    |
| Filtro a dischi in acciaio 4"<br>Steel disc filter 4"    |
| Filtro meccanico a Y 2"<br>Y -strainer 2"                |
| Filtro meccanico a Y 3"<br>Y -strainer 3"                |
| Filtro meccanico a Y 4"<br>Y -strainer 4"                |
| Filtro meccanico a Y 5"<br>Y -strainer 5"                |
| Filtro meccanico a Y 6"<br>Y -strainer 6"                |
| Valvola a 3 vie 2" miscelatrice<br>3-way mixing valve 2" |
| Valvola a 3 vie 3" miscelatrice<br>3-way mixing valve 3" |
| Valvola a 3 vie 4" miscelatrice<br>3-way mixing valve 4" |
| Valvola a 3 vie 5" miscelatrice<br>3-way mixing valve 5" |
| Valvola a 3 vie 6" miscelatrice<br>3-way mixing valve 6" |
| Valvola a 3 vie 2" on/off<br>3-way valve 2" on/off       |
| Valvola a 3 vie 3" on/off<br>3-way valve 3" on/off       |
| Valvola a 3 vie 4" on/off<br>3-way valve 4" on/off       |
| Valvola a 3 vie 5" on/off<br>3-way valve 5" on/off       |

# Optional — 2/2

## Options

PROS6-A1640-OSB · 56 optional disponibili per questa configurazione · 56 options available for this configuration

### Valvole e filtri / Valves and filters

Valvola a 3 vie 6" on/off  
3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori  
Compressor acoustic enclosure

Compressore inverter  
Inverter compressor

Desurriscaldatore a piastre  
Plate desuperheater

Rilevatore fughe gas  
Refrigerant leak detector

### Free cooling / Free cooling

Free cooling 1200 kW – 24 ventilatori (rif. 18)  
Free cooling 1200 kW – 24 fans (ref. 18)

### Ventilazione e clima / Fans and climate

Inverter ventilatori  
Fan inverter

Ventilatori EC  
EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)  
Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)  
RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)  
RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete  
Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C  
Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla  
Spring anti-vibration mounts

Antivibranti in gomma  
Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

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Continuity by design